HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered a horizontally scalable **fetch scheduler** for the data plane, designed to process up to **85 million requests per day**, enhancing the command-and-control service.
- Developed the **Fleet Manager control plane**, including provider adapters for inventory and health monitoring, configuration reconciliation, and software rollout control for a fleet of thousands of residential devices.
- Owned and enhanced the **CONNECT proxy gateway**, managing edge session lifecycles, IP reputation, and request scheduling to improve proxy service reliability.
- Implemented secure and reliable networking between control services and residential peers using **mTLS** and **gRPC** over potentially unreliable connections, addressing NAT traversal challenges.
- Defined and maintained **service-level objectives** (SLOs) for availability and request success rates across the Edge network infrastructure, monitoring peer availability and device drop-off.
- Built and deployed **Python automation scripts** for data ETL and scheduled reporting tasks, processing data from various enterprise systems.
- Containerized Java Spring Boot microservices using **Docker** and deployed them to **AWS EC2** instances, managing CI/CD pipelines with **Jenkins** for automated builds and deployments.

SKILLS

Languages – Rust, Python, Java, C++, SQL

Networking & Protocols – gRPC, mTLS, HTTP CONNECT, TCP, TLS, Protocol Buffers

Infrastructure & Cloud – AWS (EC2, S3, Lambda, CloudWatch), GCP, Kubernetes, Docker, Terraform, Oracle Cloud Infrastructure (OCI)

Distributed Systems – Schedulers, Control Planes, Stateful Services, Consistency, Backpressure, Idempotency, NAT Traversal

Proxy & Edge – Forward Proxies, Proxy Gateways, Fleet Management, Device Control, IP Reputation, Session Management

Observability & Operations – Prometheus, Grafana, Loki, OpenTelemetry, SLOs, Error Budgets, Incident Response, On-call Rotation

Data & Messaging – BigQuery, Valkey, Confluent Kafka, PostgreSQL, MySQL, Oracle

PROJECTS

Distributed Device Fleet Management System

- Designed and implemented a **control plane** for managing a large fleet of distributed devices across Linux and Windows environments, focusing on inventory, health, and configuration.
- Engineered a scalable **scheduling platform** using **Rust** and **gRPC** to manage device tasks and data extraction requests, handling over **50 million requests daily**.
- Developed provider adapters to standardize the onboarding of new device sources, reducing bespoke scripting needs and improving fleet extensibility.

Edge Networking and Proxy Service Enhancement

- Improved the **CONNECT proxy gateway** service, focusing on edge session management, IP reputation tracking, and request routing for residential web data extraction.
- Implemented robust networking solutions for communication between control services and peers over unreliable connections, integrating **mTLS** and addressing **NAT traversal**.
- Built observability tooling to monitor peer availability, device health, and bandwidth usage, contributing to improved overall system reliability and performance.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)